Sheet 47 Mon 8 Mar. 2027 *Reciprocal function: variations; relative position of $y = x$ and $y = x^2$*

1. Warm-up · Quick [CAL] Calculate $\frac{1}{2}$, $\frac{1}{-4}$ and $\frac{1}{0.1}$.

   Ans. 0.5; -0.25; 10.

2. Core · Method [CMP] Compare $\frac{1}{3}$ and $\frac{1}{5}$; what can be said about the variations of the reciprocal function on $]0;+\infty[$?

   Ans. $\frac{1}{3} > \frac{1}{5}$ although $3 < 5$: the reciprocal function is decreasing on $]0;+\infty[$ (and on $]-\infty;0[$, but not on $\mathbb{R}^*$).

3. Core + · True/false [CMP] True or false? Justify. For every real number $x > 0$, $x^2 \geqslant x$.

   Ans. False. Counterexample: $x = \frac{1}{2}$, $x^2 = \frac{1}{4} < \frac{1}{2}$. In fact $x^2 \geqslant x$ if and only if $x \geqslant 1$ (for $x > 0$).

4. Review · Quick [PER] What are the front view and the top view of a cylinder standing on its base?

   Ans. Front: a rectangle; top: a disc.

5. Thread · Prove [LIT] Show that the product of two even integers is a multiple of 4.

   Ans. With $a = 2p$, $b = 2q$: $ab = 4pq$, a multiple of 4.

Sheet 48 Tue 9 Mar. 2027 *Comparing $f(a)$ and $f(b)$; optimisation problems*

1. Warm-up · Quick [CMP] $f(x) = x^2$. Compare $f(-3)$ and $f(2)$.

   Ans. $f(-3) = 9 > 4 = f(2)$.

2. Core · Method [REG] $f(x) = x^2$, $a = -1.5$ and $b = -2$. Compare $f(a)$ and $f(b)$ without calculating, using the variations.

   Ans. $-2 < -1.5 \leqslant 0$ and f is decreasing on $]-\infty;0]$: $f(-2) > f(-1.5)$. Check: $4 > 2.25$.

3. Core + · Optimisation [MOD] $A(x) = x(10-x)$ for $x \in [0;10]$ (area of a rectangle with perimeter 20). Show that $A(x) = 25 - (x-5)^2$ and deduce the maximum.

   Ans. $25 - (x-5)^2 = 25 - x^2 + 10x - 25 = x(10-x)$; maximum 25 reached for $x = 5$ (square of side 5).

4. Review · Quick [STA] 18 pupils out of 24 have an average above 10. Give the relative frequency as a percentage.

   Ans. $\frac{18}{24} = 75\,\%$.

5. Thread · MCQ [EQ] The solution of $\frac{x}{3} + 1 = \frac{7}{2}$ is:

   A. $\frac{15}{2}$ **B.** $\frac{9}{2}$ **C.** $\frac{21}{2}$ **D.** $\frac{5}{6}$

   Ans. A. $\frac{15}{2}$

   Pitfall: forgetting to subtract the 1 before multiplying by 3 ($\frac{21}{2}$).

Cross-tabulating two qualitative variables

Sheet 49 Thu 11 Mar. 2027 *Cross-tabulating two qualitative variables*

1. Warm-up · Quick [PRO] Express $\frac{24}{60}$ as a percentage.

   Ans. 40 %.

2. Core · Quick [LEC] In a class of 30 pupils, 18 are girls. What is the relative frequency of girls? of boys?

   Ans. $\frac{18}{30} = 60\,\%$; 40 %.

3. Core + · Classify [STA] Classify each variable: eye colour; baccalauréat grade (AB, B, TB); height in cm.

   Ans. Nominal qualitative; ordinal qualitative (natural order); quantitative.

4. Review · Problem [MOD] A rectangle of perimeter 20 cm has length x. Express its area $A(x)$; calculate

$A(3)$ and $A(5)$.

Ans. Width $10 - x$; $A(x) = x(10 - x)$; $A(3) = 21$, $A(5) = 25$ cm^2.

5. Thread · Quick [SOL] Calculate the exact volume of a cylinder of radius 2 cm and height 5 cm.

Ans. $V = \pi \times 4 \times 5 = 20\pi$ cm^3.

Sheet 50 Mon 15 Mar. 2027 *Qualitative variables; two-way frequency table*

1. Warm-up · Quick [CAL] Complete: $12 + \ldots = 30$ and $22 + \ldots = 40$.

Ans. 18; 18.

2. Core · Table [LEC] Total 40 pupils. Girls: 22; canteen users (C): 25; girls who use the canteen: 15. Complete the two-way table.

	C	Not C	Total
Girls	15	…	22
Boys	…	…	…
Total	25	…	40

Ans. Girls not using the canteen 7; boys using the canteen 10; boys not using the canteen 8; total boys 18; total not using the canteen 15 (check: $7 + 8 = 15$).

3. Core + · True/false [LEC] True or false? Justify. "Satisfaction level (low, medium, high)" is a nominal qualitative variable.

Ans. False. There is a natural order: it is an ordinal qualitative variable.

4. Review · Problem [CMP] Discount A: $-30\,\%$ on 80 000 Ar. Discount B: $-20\,\%$ then $-10\,\%$. Which is the better deal?

Ans. A: 56 000 Ar; B: $80\,000 \times 0.8 \times 0.9 = 57\,600$ Ar. A is the better deal.

5. Thread · Method [DEM] We know that $AB^2 + AC^2 = BC^2$. What conclusion can be drawn? Which property is used?

Ans. ABC is right-angled at A: converse of Pythagoras' theorem.

Sheet 51 Tue 16 Mar. 2027 *Marginal frequencies*

1. Warm-up · Quick [PRO] Express $\dfrac{25}{40}$ as a percentage.

Ans. 62.5 %.

2. Core · Method [LEC] Using the table from sheet 50: calculate the marginal frequency of girls and that of canteen users.

Ans. Girls: $\dfrac{22}{40} = 55\,\%$; canteen: $\dfrac{25}{40} = 62.5\,\%$ (marginal counts over the total count).
Marginal frequency = row (or column) total / overall total.

3. Core + · Quick [LEC] The marginal frequency of girls is 55 %. What is that of boys?

Ans. 45 %: the marginal frequencies of a variable add up to 100 %.

4. Review · Quick [SUB] Calculate $(2a - b)^2$ for $a = 3$ and $b = -2$.

Ans. $(6 + 2)^2 = 64$.

5. Thread · Quick [REP] $A(-1;2)$ and $B(3;2)$. What can be said about the line (AB)? Give AB.

Ans. Horizontal (same ordinate), parallel to the horizontal axis; $AB = 3 - (-1) = 4$.

Sheet 52 Thu 18 Mar. 2027 *Conditional frequencies*

1. Warm-up · Quick [PRO] Give $\frac{15}{22}$ as a percentage rounded to the nearest tenth.

   Ans. 68.2 %.

2. Core · Method [LEC] With the same table: among the girls, what is the frequency of canteen users? Among the boys?

   Ans. Girls: $\frac{15}{22} \approx 68.2\,\%$; boys: $\frac{10}{18} \approx 55.6\,\%$ (divide by the total of the subpopulation).

3. Core + · Distinguish [LEC] Distinguish between "the frequency of girls among canteen users" and "the frequency of canteen users among girls".

   Ans. $\frac{15}{25} = 60\,\%$ and $\frac{15}{22} \approx 68.2\,\%$: the reference total is not the same.

4. Review · Construction [CON] Draw $[AB]$ of length 6 cm, then the circles with centres A and B and radius 4 cm, which meet at C and D. What type of quadrilateral is $ACBD$?

   Ans. A rhombus: $AC = CB = BD = DA = 4$ cm (four equal sides).

5. Thread · Quick [DIV] Is 91 prime?

   Ans. No: $91 = 7 \times 13$.

   Pitfall: it is divisible by none of 2, 3, 5.

Sheet 53 Mon 22 Mar. 2027 *Completing a two-way table*

1. Warm-up · Quick [PRO] Calculate 35 % of 80.

   Ans. 28.

2. Core · Method [LEC] 200 customers: 60 % are women; 25 % of the women and 50 % of the men buy online. Complete the two-way table.

   Ans. Women 120, of whom 30 online; men 80, of whom 40 online; online in total: 70.

   To go from conditional frequencies to counts, multiply by the size of the subpopulation.

3. Core + · Check [LEC] A table states "row total: $45 + 55 = 100$" but "column total: $60 + 30 = 90$". What can be concluded?

   Ans. The table is inconsistent: the grand total must be the same by rows and by columns; there is a data-entry or calculation error.

4. Review · True/false [PRB] True or false? Justify. After three consecutive "heads", the probability of getting "tails" on the next toss is greater than $\frac{1}{2}$.

   Ans. False. The tosses are independent: the probability remains $\frac{1}{2}$.

5. Thread · Quick [GEO] ABC is right-angled at A, $BC = 10$ cm, $\widehat{ABC} = 30°$. Calculate AC.

   Ans. $AC = BC \times \sin 30° = 10 \times 0.5 = 5$ cm.

   Use the sine: opposite side / hypotenuse.

Sheet 54 Tue 23 Mar. 2027 *Real data: processing a file and filtering (AND, OR, NOT)*

1. Warm-up · Quick [LEC] In a file of 1 000 rows, 400 satisfy A, 300 satisfy B and 100 satisfy both A and B. How many satisfy "A AND B"? "A OR B"?

   Ans. 100; $400 + 300 - 100 = 600$ (the intersection must not be counted twice).

2. Core · Method [LEC] With the same data: how many satisfy "NOT A"? "A AND NOT B"?

   Ans. $1\,000 - 400 = 600$; $400 - 100 = 300$.

3. Core + · Reading code [ALG] What is the final value of `compte`?

```
compte = 0
for age in [15, 16, 17, 16, 18]:
    if age >= 16 and age < 18:
        compte = compte + 1
```